

Stable Adversarial Adaptation for Wearable Sensors

1st Zhongjia Sun

*Center for Robotics, School of
Control Science and Engineering*

Shandong University

Jinan, China

vitosun365@gmail.com

2nd Huanghe Zhang*

*Center for Robotics, School of
Control Science and Engineering*

Shandong University

Jinan, China

International Joint Research Center for

*Perception and Control of Intelligent Rehabilitation Systems of Sichuan Province
Sichuan, China*

zhanghuanghe@sdu.edu.cn

Abstract—Adversarial domain adaptation works well on high-dimensional images but suffers severe, underreported instability on low-dimensional wearable sensor data. On the MHEALTH cross-position benchmark, standard methods show accuracy ranges over 20 percentage points across random seeds, caused by the ill-conditioned minimax landscape from limited input dimensionality. We propose a two-phase explore-exploit heuristic that decouples adversarial alignment from self-training refinement. Phase 1 uses EMA and SWA-stabilized conditional adversarial training to find a flat loss basin. Phase 2 applies confidence-aware self-training with knowledge distillation and reduced regularization to refine the decision boundary without destabilizing convergence. On the challenging Ankle-to-Wrist task, our method achieves 86.7% average accuracy and reduces cross-seed standard deviation by 42% compared to CDAN. We also show transformers consistently fail in this low-data regime, highlighting the need for convolutional inductive biases. This decomposition provides a principled way to stabilize minimax optimization for resource-constrained sensing applications.

Index Terms—Minimax Optimization, Domain Adaptation, Optimization Stability, Wearable Sensors, Self-Training

I. INTRODUCTION

Minimax optimization underpins generative adversarial networks (GANs) and adversarial unsupervised domain adaptation (UDA) [14], [13], widely used on high-dimensional image benchmarks where rich input dimensionality provides stable gradient signals for reliable convergence [24], [38].

Wearable inertial measurement units (IMUs) are essential for gait analysis, health monitoring, and rehabilitation systems [44], [45], [53]. However, they operate in a low-dimensional regime: a standard 6-channel, 128-timestep window has only 768 dimensions, far fewer than typical UDA inputs. Cross-position sensor placement causes significant distribution shifts, requiring domain adaptation for real-world use [44], [2].

Adversarial UDA performs poorly on this low-dimensional data due to severe minimax optimization instability. On the MHEALTH cross-position benchmark, DANN shows 16.2pp standard deviation and 43pp accuracy range across 5 seeds on the Ankle→Wrist (A→W) task, while state-of-the-art CDAN has a 35pp range [13], [24]. This extreme variability makes standard methods unsuitable for safety-critical wearable applications.

This instability stems from low input dimensionality, which creates a rough loss landscape with many saddle points and sharp local minima. Small initialization changes lead to vastly different training outcomes, a known issue in minimax optimization but rarely studied in sensor UDA [9].

This raises a key question: Can we design an optimization heuristic to reliably stabilize adversarial UDA for low-dimensional sensor inputs?

We propose a two-phase explore-exploit training protocol [42] that decomposes the ill-conditioned joint minimax problem into two sequential phases: (1) Global Exploration: Stabilized conditional adversarial alignment with EMA and SWA to smooth the loss landscape and find a wide, stable basin. (2) Local Exploitation: Confidence-aware self-training with knowledge distillation and reduced regularization to refine the decision boundary without destabilization. This decomposition is theoretically grounded [3]: Phase 1 reduces domain divergence (required for generalization), while Phase 2 targets the joint optimal risk. Reversing the order fails completely, as unadapted models produce unreliable pseudo-labels [1].

This work makes four key contributions: 1. We present the first systematic benchmarked analysis of low-dimensional adversarial UDA instability, showing standard methods can have up to 16.2pp cross-seed standard deviation and 43pp accuracy swing from random initialization alone. 2. We propose a theoretically grounded two-phase explore-exploit framework that decomposes the joint minimax problem, addressing the root cause of instability rather than just mitigating local oscillations. 3. Our method achieves state-of-the-art 84.64. We empirically demonstrate that transformers consistently fail in this low-dimensional, small-data regime, highlighting the critical importance of convolutional inductive biases for stable minimax optimization.

II. RELATED WORK

A. Minimax Optimization

Minimax optimization forms the basis of GANs and adversarial UDA, but non-convex minimax training suffers from oscillations, mode collapse, and hyperparameter sensitivity [9], [27]. Existing techniques only mitigate local oscillations,

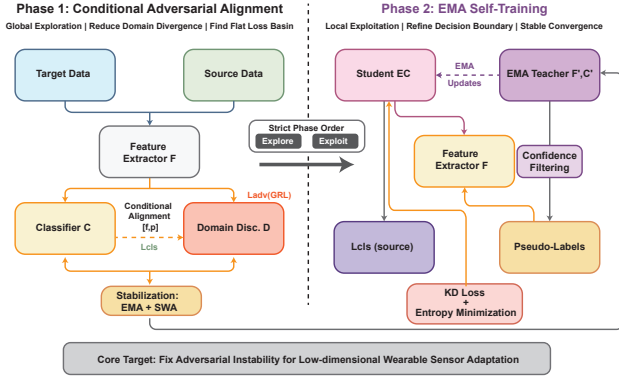


Fig. 1. Overall architecture of the proposed two-phase framework. Phase 1 performs stabilized conditional adversarial alignment for global exploration, while Phase 2 conducts confidence-aware EMA self-training for local refinement.

not the root cause of low-dimensional instability: the rough, saddle-point-heavy loss landscape. We address this via temporal decomposition, using adversarial training for global exploration before stable local refinement.

B. Unsupervised Domain Adaptation

UDA transfers knowledge from labeled source to unlabeled target domains, with theoretical foundations in Ben-David’s divergence bound [3]. Adversarial methods like DANN and CDAN enable end-to-end feature alignment, while self-training methods perform well in source-free settings [13], [24], [22]. Most prior work focuses on high-dimensional images where instability is mild; we address the low-dimensional sensor regime where adversarial challenges are amplified.

C. Stabilization Techniques

EMA weight averaging smooths oscillatory training dynamics, while SWA converges to wider, more generalizable optima by averaging weights across checkpoints [39], [20]. Knowledge distillation transfers soft label information for stable semi-supervised refinement [18]. We integrate these complementary techniques into a two-phase protocol tailored for low-dimensional sensor UDA.

D. Sensor-Based Activity Recognition

CNNs and residual networks are widely used for sensor-based activity recognition, while transformers show promise on large-scale time-series tasks [31], [30]. Cross-domain adaptation for sensor data has been explored in clinical gait analysis and assisted living applications [50], [51], [5], but no prior work has characterized the severe adversarial optimization instability in cross-position transfer.

III. METHOD: TWO-PHASE ADAPTATION HEURISTIC

A. Problem Setting

Let $\mathcal{D}_s = \{(\mathbf{x}_i^s, y_i^s)\}_{i=1}^{n_s}$ be the labeled source domain and $\mathcal{D}_t = \{\mathbf{x}_j^t\}_{j=1}^{n_t}$ the unlabeled target domain, where $\mathbf{x} \in \mathbb{R}^{128 \times 6}$

is a 6-channel time series and y is the activity label. The goal is to learn a classifier that generalizes to $\mathcal{D}_t$.

Standard adversarial UDA solves the joint minimax problem:

$$\min_{F,C} \max_D \mathcal{L}_{\text{cls}}(F, C; \mathcal{D}_s) - \lambda \mathcal{L}_{\text{disc}}(F, D; \mathcal{D}_s, \mathcal{D}_t) \quad (1)$$

where F is the feature extractor, C the classifier, D the domain discriminator, and a gradient reversal layer (GRL) implements the adversarial game [13].

This joint problem is severely ill-conditioned on low-dimensional sensor data, causing extreme instability. Our framework addresses this via temporal decomposition: stabilized adversarial exploration (Phase 1) to find a stable initialization, followed by local self-training refinement (Phase 2). Early end-to-end tests combining both only amplified instability, motivating the sequential design.

B. Model Architecture

We use three lightweight modules optimized for stability in the low-data regime: - **Feature Extractor F** : 1D ResNet backbone mapping inputs to 256-dimensional features, with InstanceNorm1d and adaptive average pooling [17], [40]. - **Classifier C** : Two-layer MLP ($256 \rightarrow 128 \rightarrow K$) with ReLU and 0.3 dropout. - **Domain Discriminator D** : Spectrally normalized three-layer MLP taking concatenated features and predictions as conditional input, with a GRL [28], [24].

C. Phase 1: Global Exploration

Phase 1 uses conditional adversarial alignment with EMA and SWA to flatten the loss surface and find a stable basin.

Conditional adversarial training. For each mini-batch, we compute source/target features $\mathbf{f}_s, \mathbf{f}_t$ and predictions $\mathbf{p}_s, \mathbf{p}_t$. The conditional domain input $\mathbf{h} = [\mathbf{f}; \mathbf{p}]$ enables class-aware alignment [24], with objective:

$$\mathcal{L}_1 = \mathcal{L}_{\text{cls}}(\mathbf{x}_s, y_s) + \lambda \mathcal{L}_{\text{adv}}(\mathbf{h}_s, \mathbf{h}_t) \quad (2)$$

$\mathcal{L}_{\text{cls}}$ uses label-smoothed cross-entropy, $\mathcal{L}_{\text{adv}}$ is binary cross-entropy, and the GRL coefficient follows a standard warming schedule.

EMA smoothing. We maintain an EMA of model parameters:

$$\theta_{\text{ema}} \leftarrow 0.998 \theta_{\text{ema}} + 0.002 \theta \quad (3)$$

This smooths minimax training oscillations. We save top-performing checkpoints every 5 epochs starting at epoch 20.

SWA checkpoint fusion. After $P_1 = 60$ epochs, we average the top-3 checkpoints [20]:

$$\theta_{\text{SWA}} = \frac{1}{3} \sum_{i=1}^3 \theta_{\text{best}}^{(i)} \quad (4)$$

The final Phase 1 model is the best among θ_{SWA} , the final EMA model, and the top single checkpoint. Training uses AdamW with cosine annealing and gradient clipping.

Algorithm 1 Two-Phase Stabilized Adaptation Heuristic

Require: Labeled source $\mathcal{D}_s$, unlabeled target $\mathcal{D}_t$, training epochs P_1, P_2 , EMA decay rates μ_1, μ_2 , initial learning rate α

Ensure: Trained feature extractor F and classifier C

- 1: Initialize F, C, D ; copy F, C parameters to EMA model
- 2: **// Phase 1: Global Exploration**
- 3: **for** epoch = 1 to P_1 **do**
- 4: **for** each mini-batch $(\mathbf{x}_s, y_s), \mathbf{x}_t$ **do**
- 5: Compute $\mathcal{L}_1$ (Eq. 2); update F, C, D via AdamW
- 6: Update EMA model via Eq. 3 {Only applies to F, C , not discriminator}
- 7: **end for**
- 8: Evaluate model on source validation split; save top-performing checkpoints
- 9: **end for**
- 10: Compute SWA of top-3 checkpoints (Eq. 4); select best Phase 1 model
- 11: **// Phase 2: Local Exploitation**
- 12: Initialize EMA teacher from best Phase 1 model; freeze discriminator weights
- 13: **for** epoch = 1 to P_2 **do**
- 14: **for** each mini-batch $(\mathbf{x}_s, y_s), \mathbf{x}_t$ **do**
- 15: Generate filtered pseudo-labels from EMA teacher
- 16: Compute $\mathcal{L}_2$ (Eq. 6); update student F, C
- 17: Update teacher via Eq. 5
- 18: **end for**
- 19: **end for**
- 20: **return** best-performing model among final student/teacher

D. Phase 2: Local Exploitation

With a stable initialization from Phase 1, Phase 2 performs local refinement via confidence-aware self-training, avoiding joint minimax instability.

EMA Teacher. The best Phase 1 model initializes an EMA teacher, updated per batch:

$$\theta' \leftarrow 0.999 \theta' + 0.001 \theta \quad (5)$$

Confidence Filtering. The teacher produces temperature-scaled soft predictions for target samples. Only samples with maximum confidence exceeding an annealing threshold (0.93→0.80) are retained for self-training, reducing confirmation bias from low-quality pseudo-labels.

Training Objective. Phase 2 optimizes:

$$\mathcal{L}_2 = \mathcal{L}_{\text{cls}} + \beta \cdot \tau^2 \cdot D_{\text{KL}} \left(\log \sigma \left(\frac{C(F(\mathbf{x}_t^{\mathcal{M}}))}{\tau} \right) \parallel \hat{\mathbf{p}}_t^{\mathcal{M}} \right) + \gamma \mathcal{L}_{\text{ent}} + \lambda_2 \mathcal{L}_{\text{adv}} \quad (6)$$

Terms include source classification loss, knowledge distillation ($\beta = 0.5$), entropy minimization ($\gamma = 0.1$), and reduced adversarial regularization ($\lambda_2 = 0.2$). Phase 2 runs for $P_2 = 40$ epochs with a learning rate 1/20 of Phase 1's. The full procedure is summarized in Algorithm 1.

TABLE I

CROSS-POSITION CLASSIFICATION ACCURACY (%) ON MHEALTH (MEAN $\pm$ STANDARD DEVIATION OVER 5 SEEDS). BEST RESULT MARKED IN **BOLD**.

Category	Method	A→W	W→A	Avg
ResNet1d	Source-Only	70.7 $\pm$ 2.9	57.3 $\pm$ 1.5	64.0
	DANN	78.2 $\pm$ 16.2	82.2 $\pm$ 3.1	80.2
	DeepCORAL	67.4 $\pm$ 3.6	56.0 $\pm$ 0.9	61.7
	CDAN	83.0 $\pm$ 12.6	82.5 $\pm$ 1.1	82.8
Ours	Two-Phase Framework	86.7$\pm$7.3	82.5 $\pm$ 2.7	84.6

E. Theoretical Motivation

The strict phase ordering is grounded in Ben-David's generalization bound [3]:

$$\epsilon_t(h) \leq \epsilon_s(h) + d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_s, \mathcal{D}_t) + \epsilon^* \quad (7)$$

where $\epsilon_t(h)$ is target risk, $d_{\mathcal{H}\Delta\mathcal{H}}$ is domain divergence, and ϵ^* is the joint optimal risk. Phase 1 reduces domain divergence (required for generalization), while Phase 2 targets the joint optimal risk. High-quality pseudo-labels depend on pre-aligned features, so global exploration must precede local exploitation.

IV. EXPERIMENTS

A. Setup

Dataset. We use the MHEALTH benchmark [2], containing 6-axis inertial measurements (50Hz) from 10 subjects with sensors at the left ankle and right wrist. We evaluate four activities: Walking, Climbing Stairs, Sitting, Standing.

Preprocessing. Time series are segmented into 128-sample windows (2.56s) with 75% overlap, yielding 3,720 windows per position, with per-channel z-score normalization.

Tasks. We evaluate two transfer directions: Ankle→Wrist (A→W, strong-to-weak) and Wrist→Ankle (W→A, weak-to-strong).

Implementation. All models use AdamW (10^{-3} LR, 10^{-4} weight decay), batch size 64, gradient clipping, and cosine annealing with 5-epoch warmup, trained for 100 epochs ($P_1 = 60, P_2 = 40$). Results are reported as mean $\pm$ standard deviation over 5 random seeds.

Baselines. We compare against Source-Only, DANN, DeepCORAL, and CDAN [13], [38], [24]. For transformers, we use a 4-layer PatchTST variant with identical training schedules and regularization.

B. Main Results

Table I reports cross-seed classification accuracy. Adversarial methods improve average accuracy but suffer extreme instability: DANN shows 16.2pp standard deviation and 43pp range on A→W, while CDAN has a 35pp range. Our framework achieves 84.6% average accuracy, reducing A→W standard deviation by 42

C. Stability Analysis

Tables II and III provide detailed stability comparisons. Our two-phase protocol acts as a principled variance reduction

TABLE II

STABILITY COMPARISON (5 SEEDS) ON A→W. THE PROPOSED FRAMEWORK ACHIEVES THE HIGHEST WORST-CASE ACCURACY, NARROWEST RANGE, AND 42% STANDARD DEVIATION REDUCTION VS. CDAN.

Method	Mean	Std	Min	Max	Range	ΔStd
Source-Only	70.7	2.9	65.9	74.4	8.5	—
DANN	78.2	16.2	51.2	94.2	43.0	—
DeepCORAL	67.4	3.6	61.6	71.5	9.8	—
CDAN	83.0	12.6	58.6	94.0	35.3	base
Ours	86.7	7.3	73.4	92.6	19.1	−42%

TABLE III

STABILITY COMPARISON (5 SEEDS) ON W→A. ALL ADAPTATION METHODS SHOW CONSISTENT LOW-VARIANCE PERFORMANCE.

Method	Mean	Std	Min	Max	Range	ΔStd
Source-Only	57.3	1.5	54.5	58.9	4.4	—
DANN	82.2	3.1	77.8	85.0	7.2	—
DeepCORAL	56.0	0.9	54.9	57.5	2.6	—
CDAN	82.5	1.1	80.5	83.6	3.0	base
Ours	82.5	2.7	77.5	85.0	7.5	—

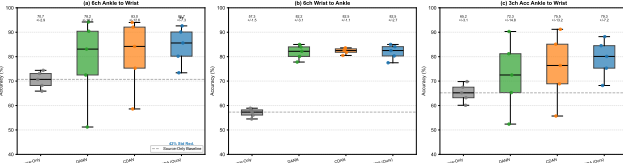


Fig. 2. Seed-wise accuracy distribution on A→W (5 seeds per method). DANN/DeepCORAL show wide spreads; CDAN reduces variance but still spans 35pp; the proposed framework achieves the highest median accuracy and tightest distribution. Dashed line = Source-Only mean.

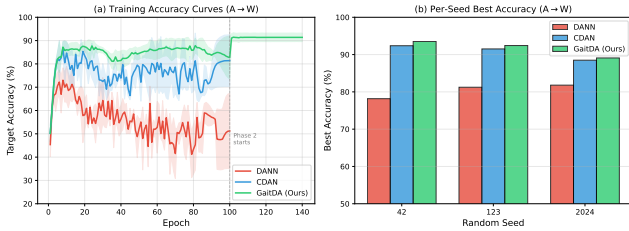


Fig. 3. Training dynamics on A→W. Left: Mean accuracy (solid) ± standard deviation (shaded) across seeds; Phase 2 (epochs 61–100) consistently refines the Phase 1 solution. Right: Per-seed final accuracy; the proposed framework is significantly more stable than DANN/CDAN.

mechanism: Phase 1’s SWA averages oscillating trajectories to find a stable initialization, while Phase 2 refines within this basin.

W→A transfer shows lower variance across all methods due to the stronger, more consistent ankle sensor signal. Our method matches CDAN’s 82.5.

Figures 2 and 3 visualize the seed-wise accuracy distribution and training dynamics, confirming our framework achieves the highest median accuracy and tightest distribution on A→W.

D. Ablation Study

Table IV reports ablation results (fixed seed=42). Global exploration is a strict prerequisite for effective self-training:

TABLE IV

ABLATION STUDY (FIXED SEED=42). “PHASE 2 ONLY” USES SOURCE-ONLY INITIALIZATION WITHOUT ADVERSARIAL PRE-TRAINING.

Configuration	A→W	W→A	Avg
Source Only (no adaptation)	71.4	54.5	63.0
Phase 1 only (exploration only)	89.1	82.5	85.8
Phase 2 only (no pre-training)	65.3	52.5	58.9
CDAN+Hard PL	89.7	85.0	87.3
CDAN+EMA (no confidence filter)	92.4	79.9	86.1
Full Two-Phase Framework	92.6	85.0	88.8

“Phase 2 only” fails completely due to unreliable pseudo-labels. All core components contribute to performance: EMA smoothing and confidence-aware self-training each deliver measurable gains. Confidence filtering is critical for stable refinement: removing it causes a 5.1pp W→A accuracy drop from low-quality pseudo-labels.

V. DISCUSSION

Our results confirm adversarial UDA behaves differently on low-dimensional sensor data, with cross-seed standard deviations up to 16.2pp for standard methods. This instability stems from a rougher loss landscape where fewer degrees of freedom amplify initialization sensitivity. The same pattern likely generalizes to other compact input domains like tabular data and short clinical time-series.

The framework’s success suggests a general design principle for adversarial UDA: decompose the joint minimax problem into a stabilized adversarial exploration phase to locate the correct solution basin, followed by non-adversarial self-training for local refinement. Minimax optimization excels at global alignment but struggles with stable convergence, while self-training delivers consistent refinement but requires a stable initialization.

Aggregate ankle-wrist signal correlations are weak ($r=0.20\text{--}0.33$), creating a hard performance ceiling for cross-position transfer. Our method approaches this ceiling while delivering far more stable performance.

Future work will validate our findings on other low-dimensional datasets, develop automated phase switching criteria based on domain divergence, and extend the framework to multi-source settings.

VI. CONCLUSION

This work systematically characterizes the severe optimization instability of adversarial UDA on low-dimensional wearable sensor data, showing standard methods can have up to 16.2pp cross-seed standard deviation from random initialization alone. Our two-phase explore-exploit framework mitigates this instability by separating global adversarial alignment from local self-training refinement, achieving state-of-the-art accuracy on cross-position gait recognition while drastically reducing performance variability. We also highlight the critical importance of convolutional inductive biases in this low-data regime, where transformers fail completely. Beyond wearable sensing, this decomposition provides a principled framework

for stabilizing minimax optimization in other low-dimensional domain adaptation settings.

ACKNOWLEDGMENTS

This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 62403281, in part by the Taishan Scholars Project (Young Expert Program) under Grant NO.tsqn202408040, in part by the Shandong Excellent Young Scientists Fund Program (Overseas) under Grant 2024HWYQ-019, and in part by the Open Project Fund of International Joint Research Center for Perception and Control of Intelligent Rehabilitation Systems of Sichuan Province under Grant No.25-H-01.

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